



Islamic University of Technology

Board Bazar, Gazipur, Dhaka

SWE 4404 – Software Project Lab-II

Project Proposal

ALUMNET

Student-Alumni Mentorship and Career Guidance Platform

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SPL Team - 11

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Chapter 1

Project Overview & Motivation

1.1 Overview

Project Name: Alumnet

This project is a web-based Student–Alumni Mentorship and Career Guidance platform designed to connect students with alumni for mentorship, networking, and career development. It provides an organized system for students to discover and connect with relevant alumni based on shared interests, career goals, and professional expertise. Through mentorship and referral features, students can seek guidance, career advice, and professional opportunities, while alumni can support students by sharing experiences, posting job or internship opportunities, and offering referrals. Additionally, the platform fosters a collaborative community through discussion forums, knowledge sharing, and event updates, creating a centralized ecosystem that supports student growth, alumni engagement, and career readiness.

1.2 Motivation

Students often struggle to find proper career guidance, mentorship, and referral opportunities despite having access to experienced alumni networks. Communication between students and alumni is usually unstructured and mostly dependent on personal connections or social media platforms, which makes meaningful interaction difficult. At the same time, many alumni are willing to help students, but there is no centralized system to efficiently connect them. This project aims to bridge that gap by creating a dedicated student–alumni networking and mentorship platform where students can connect with relevant alumni, explore career paths, and access opportunities in a more organized and collaborative way.

Chapter 2

User Requirement Analysis

2.1 Functional Requirements

Student Requirements

1. Students shall be able to create and manage their accounts securely.
2. Students shall be able to create and update profiles containing skills, interests, and career goals.
3. Students shall be able to search and filter alumni based on attributes such as company, role, and skills.
4. Students shall be able to send mentorship and referral requests to alumni.
5. Students shall be able to communicate with alumni through real-time chat.
6. Students shall be able to access career roadmaps, internship postings, and job opportunities.
7. Students shall be able to participate in discussions, forums, and events.

Alumni Requirements

1. Alumni shall be able to create and manage professional profiles.
2. Alumni shall be able to accept or reject mentorship and referral requests.
3. Alumni shall be able to share career advice, experiences, internships, and job opportunities through posts.
4. Alumni shall be able to communicate with students through real-time chat.
5. Alumni shall be able to participate in forums, discussions, and events.

Admin Requirements

1. Admin shall be able to verify student and alumni identities.
2. Admins shall be able to ban or restrict users who violate platform policies.

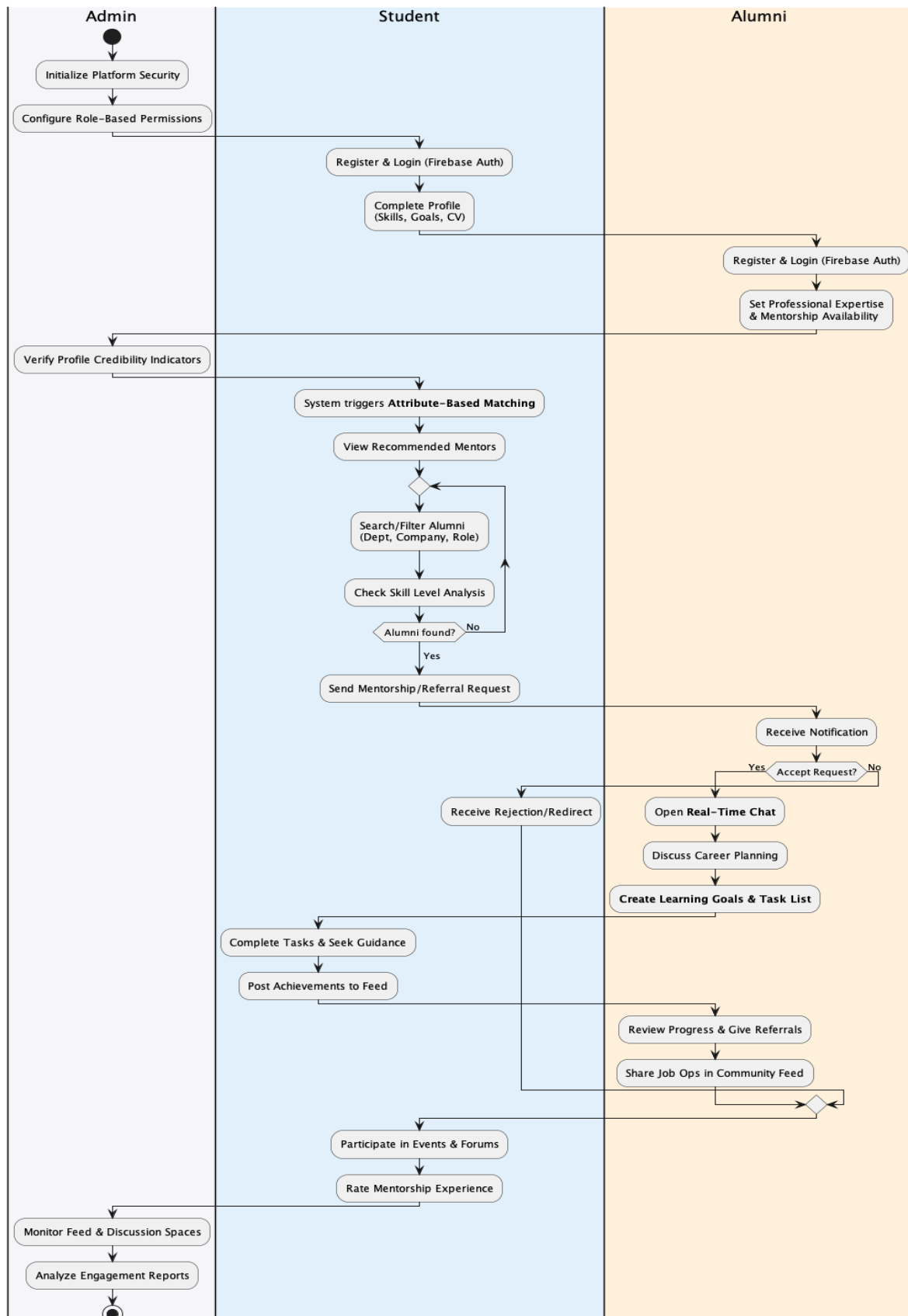
3. Admins shall be able to remove spam or inappropriate posts and comments.
4. Admins shall be able to monitor platform statistics and user activities.
5. Admins shall be able to manage events, reports, and overall platform operations.

2.2 Non-Functional Requirements

1. The system shall provide secure authentication and protected access for all users.
2. The platform shall support real-time communication with minimal delay.
3. The system shall maintain data consistency and reliability using a relational database.
4. The application shall provide a responsive and user-friendly interface across different devices.
5. The system shall be scalable to support increasing numbers of users and interactions.
6. The platform shall ensure efficient performance for searching, filtering, and loading data.
7. The system shall maintain availability and stability during regular platform usage.
8. The application shall follow a modular and maintainable architecture for future expansion and updates.

Chapter 3

Flow Chart



Chapter 4

Key Features

4.1 User Authentication & Authorization

The platform will provide a secure authentication system for both students and alumni. Users will be able to register, log in, and manage their accounts safely using Firebase Authentication. Role-based authorization will ensure that users can only access features and actions permitted for their account type.

Use Case: A student or alumnus registers on the platform, logs in securely, and accesses features based on their assigned role.

4.2 User Profile System

Each user will have a personalized profile containing information such as academic background, skills, career interests, experience, and social links. Students can showcase their goals and interests, while alumni can share their professional journey and expertise to help students connect with suitable mentors.

Use Case: A student creates a profile with skills and career goals, while an alumni member updates professional information to help students connect with suitable mentors.

4.3 Alumni Search & Filtering

Students will be able to search and filter alumni based on different criteria such as department, company, job role, skills, and areas of expertise. This feature will help students quickly find alumni who match their career interests and mentorship needs.

Use Case: A student searches for alumni working in a specific company or field to receive career guidance and mentorship.

4.4 Attribute-Based Mentor Matching

The platform will suggest suitable mentors to students based on shared attributes such as career goals, skills, interests, and preferred fields. This matching system will make it easier for students to discover alumni who can provide relevant guidance and support.

Use Case: The platform recommends alumni mentors to students based on shared interests, skills, and career aspirations.

4.5 Mentorship & Referral System

Students can send mentorship or referral requests to alumni through the platform. Alumni will be able to accept or reject requests and manage ongoing mentorship connections. This feature will help students receive career advice, interview preparation support, and professional referrals. Mentors will be able to create customized task lists and learning goals for mentees after mentorship sessions, allowing structured guidance and progress tracking throughout the mentorship journey.

Use Case: A student sends a mentorship request to an alumni mentor, receives guidance, and follows a personalized roadmap or task list created by the mentor.

4.6 Post & Feed System

Users will be able to create and share posts related to career advice, achievements, opportunities, and experiences. A centralized feed system will allow students and alumni to stay updated with community activities and shared knowledge.

Use Case: Alumni share internship opportunities or career advice through posts, while students interact through likes and comments.

4.7 Real-Time Chat System

The platform will include a real-time chat system that allows direct communication between students and alumni. Users can discuss

mentorship topics, career guidance, and professional opportunities instantly through one-to-one messaging.

Use Case: A student communicates directly with an alumni mentor to discuss interview preparation, career planning, or project guidance.

4.8 Community & Events

The platform will support community engagement through discussion spaces and event updates. Users can participate in discussions, share knowledge, and stay informed about seminars, workshops, networking sessions, and other career-related events.

Use Case: Users participate in discussion forums and register for workshops, seminars, and networking events hosted through the platform.

4.9 Additional Features

- **Skill Level Analysis:** User skills can be analyzed based on profile information, projects, certifications, and experience to better represent expertise and career readiness
- **Profile Credibility Indicators:** Verified achievements and contributions can be highlighted to help students identify reliable mentors and professionals

Use Case: The platform highlights verified achievements and analyzes user skills to improve mentor credibility and student career readiness.

Chapter 5

Tools & Technologies

5.1 Programming Language

We are using **JavaScript** as the primary programming language for both frontend and backend development. Using a single language across the full

stack ensures consistency, simplifies development, and improves team productivity.

5.2 Frontend

For frontend development, we are using **Next.js** to build a scalable and responsive web application with efficient routing and modern UI architecture. **Tailwind CSS** is used for designing clean, responsive, and customizable user interfaces quickly. Additionally, **Framer Motion** is used to add smooth animations and interactive transitions, enhancing the overall user experience.

5.3 Backend

For backend development, we are using **Node.js** as the runtime environment to build a fast and efficient server-side system. **Express.js** is used as the backend framework to handle API development, routing, middleware, and server-side logic, enabling structured communication between the frontend and database.

5.4 Database

We are using **PostgreSQL** as the primary relational database for storing structured application data securely and efficiently. **Prisma** is used as the ORM (Object Relational Mapper) to simplify database operations, schema management, and interaction between the backend and PostgreSQL.

5.5 Version Control & IDE

We are using **Git** for version control to track changes in our codebase and manage contributions from all team members. Additionally, **GitHub** serves as our remote repository, which allows us to collaborate, review code, and maintain an organized development workflow.

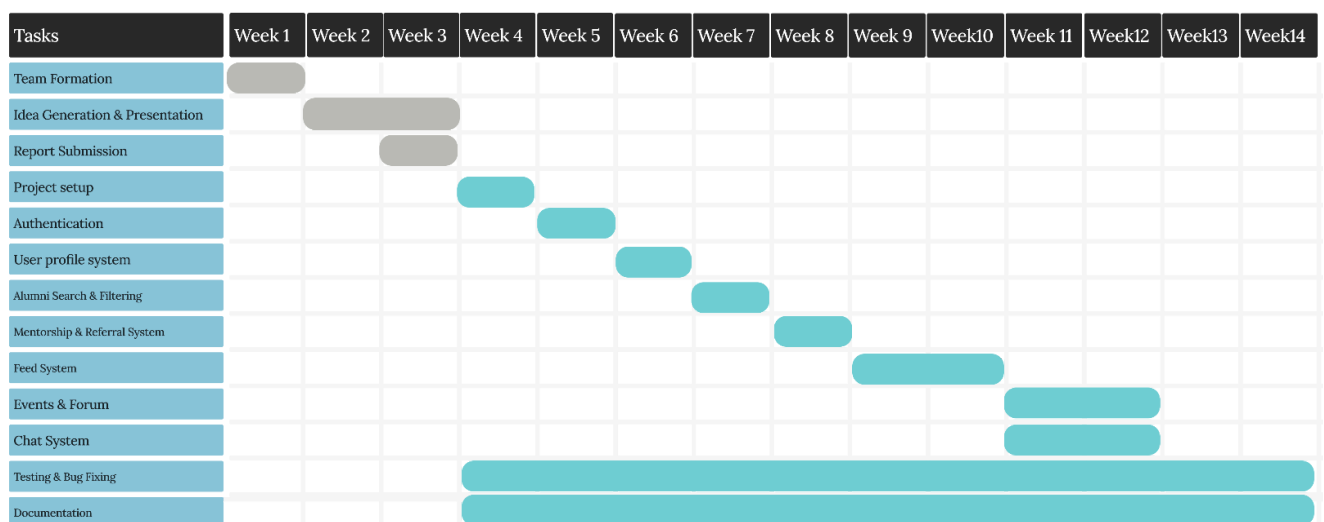
For development, we are using **Visual Studio Code (VS Code)** as our primary Integrated Development Environment (IDE)

5.6 Other Tools & Technologies

We are using **Firebase** for authentication and real-time chat functionality, enabling secure user login and instant one-to-one communication between students and alumni. **Redis** is used for caching and performance optimization to improve system speed and efficiency. **Axios** is used for handling API requests between the frontend and backend, ensuring smooth data communication throughout the platform.

Chapter 6

Proposed Timeline



Chapter 7

Action Taken based on the Feedback

Implementation of Extra Features:

In the Proposal Presentation, we were told to add some extra features apart from the existing ones.

Actions Taken:

Personalized Mentorship Roadmap System:

Mentors will be able to create customized task lists and learning goals for

mentees after mentorship sessions, allowing structured guidance and progress tracking throughout the mentorship journey.

We've also added additional features such as the **Skill Level Analysis & Profile Credibility Indicators**.